For Meeting Sept 24, 1958 Concerning Westinghouse Proposal

1. Wash, Saul Gass
Midwest: 2 Pittsburg: ~ 9:30
2. Rip Miller
3. Ken Powell . appl sci Pitts
4. Bob deSio on wed
5. Eastern: Bill McClland on wed
6. Frank Fisher - salesman
7. Lemieux - solar appl. sci
8. also Joe Sheehan
9. Kobsky

at Meeting Sept 24, 1958

F. A. Fisher	G.E.	
Saul Gass	Wash Fed	
D. Mills	W	
K. Powell	W	
J. A. Sheehan	DPHQ	370 White Plains
P. W. DeSio	Chicago Region	
G. Lemieux	GE	
W. M. Clelland	Eastern Region	
J. A. Sheehan	?	
E W Coffin		
JGK		

Westinghouse:

6 ft high × 6 ft dia

need at least 50 pts along radius. 200 × 200 pts better was done on NORC.

for 3D control rod calcs. { 3000 to 6000 pts in Plane; 20 to 50 planes } would be 1500 hrs on 704. (eg. 50 × 50 × 50) × N_g

Local reactivity can drop 50% ℓ in Time

Presently using "few group diffusion Theory" — would prefer transport

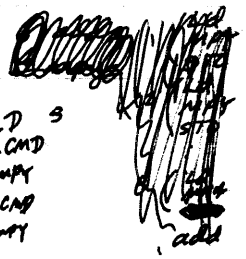
No. of iterations: 100 typical (250-500 sometimes)

50-100 outer (source iters)

For 50 × 50 points: 8 sec on 704 $\frac{8\times10^6\,\mu s}{(2.5\times10^3\ pts)(80\,\mu s/op)} = (0.4)\,10^2 \doteq 40\ ops$ per point

Storage: N values of ϕ
N values of Q
4N values of ϕ weighting } = 6N

3D basic loop:



1 (Ld, mpy, add, sto) 5

2 (Ld, mpy, add, sto)

⋮

8 (Ld, mpy, add, sto)

or

1 (LD, LCMD, CMPY) 3

2 (LCMD, CMPY)

⋮

8 (LCMD, CMPY, STO)

Simplest calc:

8 Mpy-adds 8 words from disk (20%)

1.2 + 0.6 = 1.8 μs + .36 + .2 = (2.36 μs)

2.5 + 1.0 = 3.5

Short Technical Proposal.

The purpose of this appendix is to set down very briefly and incompletely the main advantages of

Meeting: Sept 24, 1958

Rip Mills Fisher Sheehan DeSio
Ken Powell Semenuk Coffin
Saul Gass McClelland Kolehy

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question: 1 or 2 Stretchs ⇒ (Joint committee does not include STRETCH only TX)

GE definitely wants a STRETCH too

question: { Combustion Engr. } have been left out
{ NYU
{ Westinghouse
{ GE

Letter by C. DeC to Westinghouse: Prices Oct 1.

Technical Proposal - cover letters are different to each co.

7090X 1 Dec '59, 1 Feb '60
STRETCH 1 let quota of 60.

Westinghouse - def wants priority - They have done most of work have most of money.
- They want price
- Then go to AEC for approval

Sigma	70,000	Printer	
Memories @	16,000		
Disk + HS. ch	22,000,		
cd Punch	4,050	~4,000	
cd Reader	4,050	~4,000	
Printer	17,500		

Guesses of system costs

SMALL		~~Medium~~ Standard		Large	
console	5,375.	console	5,375.	console	5,325.
CPU	70,000.	CPU	70,000	CPU	70,000
2 Mems	32,000.	6 Mems	96,000	16 Mems	256,000.
Cd Rdr	4,000 ?	Cd Rdr	4,000 ?	Cd Rdr	4,000
Cd Punch	4,000 ?	Cd Punch	4,000 ?	Cd Punch	4,000
Printer	17,500 ?	Printer	17,500 ?	Printer	17,500.
4 tapes	4,400.	8 tapes	8,800.	130 tapes	33,000.
2 tape control.	3,250	2 tape Control	3,250	4 tcus	6,800.
		1 Disk	21,150.	2 Disks	26,400.
	~ 140,000.		230,000		422,750.

5375
5250
22150

8
17.5
3.6
3.25
37.35

STANDARD

			price
1.	(CPU and console)	70,000.	75,000.
2.	6 Memories	73,000	96,000
3.	1 Disk & HS Exchange	20,000.	21,000
		123,000.	192,000
4.	Cd Rdr, Cd Punch, Printer, 2 tape control units, 8 tapes	~ 37,000.	37,000
		210,000.	429,000

GE	WEST	IDTMB
7090	704	704
709TX	709TX + 704	
	STRETCH	

4 Machines?

Machine: — Space requirements
Maintenance - reliability
— Machine Features, ← main ideas

list of components,

machine building blocks constructing
input-output

problem — other general purpose machine
— good for certain diff. cases,

contrast 10 main differences

concentrate mainly on applications of Beta Chem,